

Movie Rental Data Warehouse Design Report

1. Introduction

This assignment focuses on designing a high-level data warehouse for a movie rental business. The original system is an OLTP database that supports daily operational activities such as managing customers, films, inventory, rentals, payments, staff, stores, and locations.

The OLTP database is suitable for transaction processing because it is designed to store and update operational data efficiently. However, it is not optimized for analytical reporting. Business managers need a data warehouse that can help them analyze rental activity, revenue, film popularity, customer behavior, store performance, staff performance, and trends over time.

The proposed data warehouse reorganizes the operational database into a dimensional model. The design uses fact tables to store measurable business events and dimension tables to provide descriptive context for analysis. Python was used to implement the ETL process, and MySQL was used to store both the source database and the data warehouse.

2. Business Questions

The data warehouse is designed to answer analytical business questions that support decision-making. These questions help managers

understand business performance from different perspectives such as films, customers, stores, staff, location, and time.

The main business questions are:

1. Which films are rented most frequently?

This helps managers identify the most popular films and make better inventory decisions.

2. Which films generate the highest revenue?

This helps the business understand which films contribute most to income.

3. Which film categories are most popular?

This supports decisions about which categories should be expanded or promoted.

4. Which stores generate the highest number of rentals?

This helps compare store performance based on rental activity.

5. Which stores generate the highest revenue?

This helps managers identify the strongest branches financially.

6. Which customers rent the most films?

This helps identify active and loyal customers.

7. Which customers generate the highest revenue?

This supports customer targeting and loyalty programs.

8. How does rental activity change over time?

This helps identify seasonal or monthly rental trends.

9. How does revenue change by month, quarter, or year?

This supports financial planning and performance monitoring.

10. Which staff members process the highest number of rentals or payments?

This helps evaluate staff productivity.

11. Which cities or countries generate the highest customer activity?

This helps understand customer demand by location.

12. What is the average rental duration for different films or categories?

This helps analyze customer rental behavior.

13. Which films are returned late most often?

This helps identify films or rental patterns associated with late returns.

14. How does store performance differ by location?

This helps compare branches based on geographic area.

15. What are the most active customer locations?

This helps marketing teams focus on high-activity locations.

3. Dimensional Model Design

3.1 Selected Business Processes

The main business processes selected from the OLTP movie rental database are:

1. Film Rental Transactions

This process records rental activities when customers rent films from a store.

2. Payment Transactions

This process records payments made by customers for rentals.

These two processes were selected because they directly support the most important business analysis areas: rental activity and revenue.

3.2 Fact Tables

3.2.1 fact_rental

The fact_rental table stores information about rental transactions.

Business Process:

Film rental transaction.

Grain:

Each row in fact_rental represents one rental transaction for one film copy rented by one customer from one store and processed by one staff member.

Primary Key:

rental_fact_key

Foreign Keys:

- rental_date_key
- return_date_key
- customer_key
- film_key
- store_key
- staff_key

Measures:

- rental_count: *used to count rental transactions.*
- rental_duration_days: *number of days between rental date and return date.*

- `expected_rental_duration`: *expected rental duration from the film table.*
- `late_return_flag`: *indicates whether the film was returned late.*

This fact table can answer questions about film popularity, customer rental activity, store rental performance, staff rental processing, rental duration, and late returns.

3.2.2 fact_payment

The fact_payment table stores information about payment transactions.

Business Process:

Payment transaction.

Grain:

Each row in fact_payment represents one payment transaction made by a customer for a rental.

Primary Key:

`payment_fact_key`

Foreign Keys:

- `payment_date_key`
- `customer_key`
- `film_key`
- `store_key`
- `staff_key`

Measures:

- `payment_amount`: *amount paid by the customer.*
- `payment_count`: *used to count payment transactions.*

This fact table can answer questions about revenue by film, customer, store, staff, and time.

3.3 Dimension Tables

dim_date

The dim_date table supports time-based analysis.

Main Attributes:

- date_key
- full_date
- day_number
- day_name
- month_number
- month_name
- quarter_number
- year_number

Source Data:

- rental.rental_date
- rental.return_date
- payment.payment_date

Purpose:

This dimension allows analysis by day, month, quarter, and year.

dim_customer

The dim_customer table stores customer information.

Main Attributes:

- customer_key
- customer_id

- customer_full_name
- email
- active_status
- address
- city
- country
- create_date

Source OLTP Tables:

- customer
- address
- city
- country

Purpose:

This dimension supports analysis of customer behavior, active customers, and customer locations.

dim_film

The dim_film table stores film information.

Main Attributes:

- film_key
- film_id
- title
- description
- release_year
- language_key
- rental_duration
- rental_rate
- length
- replacement_cost
- rating

Source OLTP Tables:

- film
- language

Purpose:

This dimension supports analysis of film popularity, revenue by film, rental duration, and late returns.

dim_language

The dim_language table stores film language information.

Main Attributes:

- language_key
- language_id
- language_name

Source OLTP Table:

- language

Purpose:

This dimension allows analysis of films by language.

dim_category

The dim_category table stores film category information.

Main Attributes:

- category_key
- category_id
- category_name

Source OLTP Tables:

- category
- film_category

Purpose:

This dimension supports analysis of rental activity and revenue by film category.

dim_actor

The dim_actor table stores actor information.

Main Attributes:

- actor_key
- actor_id
- actor_full_name

Source OLTP Tables:

- actor
- film_actor

Purpose:

This dimension supports analysis of film performance based on actors.

dim_store

The dim_store table stores store information.

Main Attributes:

- store_key
- store_id
- manager_staff_id
- address
- city
- country

Source OLTP Tables:

- store

- address
- city
- country

Purpose:

This dimension supports analysis of store performance by rentals, revenue, and location.

dim_staff

The dim_staff table stores staff information.

Main Attributes:

- staff_key
- staff_id
- staff_full_name
- email
- active_status
- store_id
- address
- city
- country

Source OLTP Tables:

- staff
- store
- address
- city
- country

Purpose:

This dimension supports analysis of staff productivity in processing rentals and payments.

3.4 Bridge Tables

bridge_film_category

This bridge table connects films with categories.

Attributes:

- film_key
- category_key

It is used because the relationship between films and categories is handled through the OLTP table film_category.

bridge_film_actor

This bridge table connects films with actors.

Attributes:

- film_key
- actor_key

It is used because the relationship between films and actors is many-to-many.

3.5 Schema Type

The proposed design uses a **hybrid star schema**.

Most dimensions are directly connected to the fact tables, which follows the star schema style. However, bridge tables are used to handle many-to-many relationships between films and categories, and between films and actors. Therefore, the design is considered a hybrid star schema.

The shared dimensions between `fact_rental` and `fact_payment` are:

- `dim_date`
- `dim_customer`
- `dim_film`
- `dim_store`
- `dim_staff`

4. Dimensional Model Diagram

The dimensional model diagram shows the relationship between the fact tables, dimension tables, and bridge tables.

The main fact tables are `fact_rental` and `fact_payment`. Both fact tables share common dimensions such as date, customer, film, store, and staff. The film dimension is also connected to language, category, and actor dimensions.

The diagram is included in the project folder:

`diagrams/dimensional_model.png`

5. ETL Design

The ETL process explains how data is moved from the OLTP source database into the data warehouse.

5.1 Extract

The data is extracted from the OLTP database `movie_rental_source`.

The main source tables are:

- `rental`: used to build rental transactions.

- *payment: used to build payment transactions and revenue measures.*
 - *customer: used to build the customer dimension.*
 - *film: used to build the film dimension.*
 - *inventory: used to connect rentals to films and stores.*
 - *store: used to build the store dimension.*
 - *staff: used to build the staff dimension.*
 - *address: used to get address information for customers, stores, and staff.*
 - *city: used to get city names.*
 - *country: used to get country names.*
 - *category: used to build the category dimension.*
 - *film_category: used to connect films with categories.*
 - *language: used to build the language dimension.*
 - *actor: used to build the actor dimension.*
 - *film_actor: used to connect films with actors.*
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5.2 Transform

Several transformations were applied before loading the data into the warehouse.

The main transformations include:

- *Joining related OLTP tables to create complete dimension records.*
- *Combining first_name and last_name into full name fields for customers, staff, and actors.*
- *Creating date_key values using the format YYYYMMDD.*
- *Removing duplicated records before loading.*

- *Standardizing active status values into readable labels such as Active and Inactive.*
 - *Converting operational IDs into warehouse surrogate keys.*
 - *Calculating rental_duration_days using the difference between rental_date and return_date.*
 - *Calculating late_return_flag by comparing the actual rental duration with the expected rental duration.*
 - *Handling missing return dates by allowing return_date_key and rental_duration_days to be null.*
 - *Handling many-to-many relationships using bridge tables for films with categories and actors.*
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5.3 Load

The transformed data is loaded into the data warehouse database movie_rental_dw.

The loading order is important because fact tables depend on dimension surrogate keys. Therefore, dimension tables are loaded before fact tables.

The loading order used in the project is:

- 1. dim_date*
- 2. dim_customer*
- 3. dim_language*
- 4. dim_film*
- 5. dim_category*
- 6. dim_actor*
- 7. dim_store*
- 8. dim_staff*
- 9. bridge_film_category*

10. bridge_film_actor
11. fact_rental
12. fact_payment

This order ensures that all dimension keys exist before loading the fact tables.

6. Data Quality Considerations

The ETL process should include data quality checks to ensure reliable analytical results.

The main data quality rules are:

1. **No duplicate dimension records**

Each customer, film, actor, category, store, staff, and language should appear only once in its dimension table.

2. **Valid foreign keys**

Every foreign key in the fact tables should reference an existing record in the related dimension table.

3. **Valid date keys**

Rental, return, and payment dates should be converted correctly into date keys.

4. **Handling missing return dates**

If a rental has not been returned yet, the return date should be allowed to remain null.

5. **Non-negative payment amounts**

Payment amounts should not be negative.

6. Valid rental duration

Rental duration should not be negative when return dates are available.

7. Consistent text values

City, country, language, and active status values should be standardized.

8. Bridge table uniqueness

The bridge tables should not contain duplicate film-category or film-actor combinations.

7. Conclusion

This project proposed and implemented a high-level data warehouse design for a movie rental OLTP database. The original operational schema was transformed into a dimensional model that supports analytical reporting.

The data warehouse includes rental and payment fact tables, shared dimensions, and bridge tables to handle many-to-many relationships. The ETL process was implemented using Python, extracting data from the OLTP database, transforming it into analytical structures, and loading it into the MySQL data warehouse.

The final design supports important business questions related to rental activity, revenue, film popularity, customer behavior, store performance, staff productivity, location-based activity, and time-based trends. This makes the data warehouse useful for business decision-making and reporting.